GPU Microarchitectural Power Attack Taxonomy

NVIDIA Jetson AGX Thor (Blackwell, sm_110, 130W TDP) | Baseline: 46.8W | INA3221 @ 19.74V

Memory Subsystem

1. Cache-Eviction Stress

83.8W

1.79×

`PTX st.global.wt / ld.global.cg`

`Bypass L2 → force DRAM access`

Compute Pipelines

2. Compute Saturation

61.7W

1.32×

`INT32 multiply-add chain`

`Loop-carried dependency`

Timing / DVFS

3. Pulsed Modulation

83.4W

1.78×

`clock64() cycle-accurate timing`

`0.5 Hz – 10 kHz, 50% duty cycle`

Warp Scheduler

4. Occupancy Flooding

70.1W

1.50×

`Max thread blocks, min resources`

`Saturate SM warp slots`

Thermal Side-Channel

5. Temp-Synchronized

83.3W

1.78×

`Pre-heat → cool → trigger on dT/dt`

`Negative result: ambient drift`

Memory + Compute

6. Combined Sustained

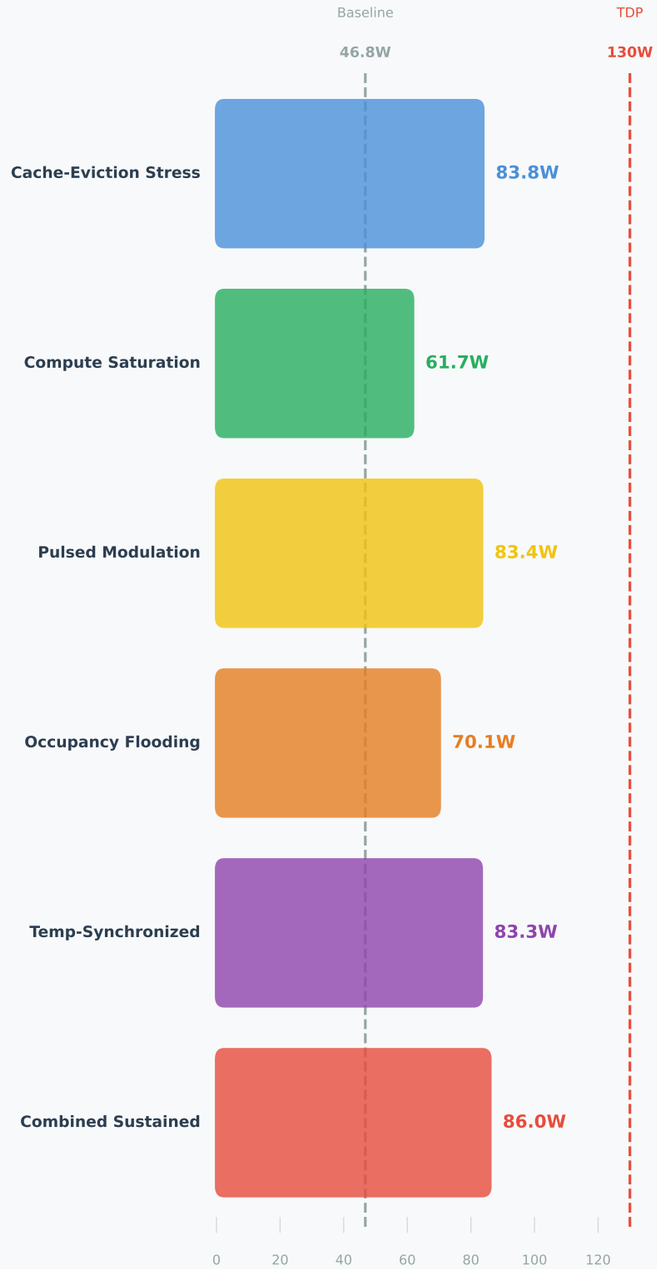
86.0W

1.83×

`Cache-eviction + compute (DC mode)`

`Peak: 105.4W (81.1% TDP)`

Measured Power (W)



Victim Impact

(Latency Degradation)

Matrix Multiply

(Compute-bound)

+92%

CNN Training

(Mixed)

+32%

CNN Inference

(Lightweight)

+4%

Most valuable targets

are the most vulnerable